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REGIONAL OFFICE SILCHAR
PRE- BOARD EXAM-2

CLASS- X (2024-25)

MATHEMATICS STANDARD (041)

TIME: 3 hours MAX.MARKS: 80

General Instructions:

Read the following instructions carefully and follow them:
1. This question paper contains 38 questions.
2. This Question Paper is divided into 5 Sections A, B, C, D and E.
3. In Section A, Questions no. 1-18 are multiple choice questions (MCQs) and questions no. 19 and 20 are Assertion- Reason based questions of 1 mark each.
4. In Section B, Questions no. 21-25 are very short answer (VSA) type questions, carrying 02 marks each.
5. In Section C, Questions no. 26-31 are short answer (SA) type questions, carrying 03 marks each.
6. In Section D, Questions no. 32-35 are long answer (LA) type questions, carrying 05 marks each.
7. In Section E, Questions no. 36-38 are case study based questions carrying 4 marks each with sub parts of the values of 1, 1 and 2 marks each respectively.
8. All Questions are compulsory. However, an internal choice in 2 Questions of section B, 2 Questions of section C and 2 Questions of section D has been provided. And internal choice has been provided in all the 2 marks questions of Section E.
9. Draw neat and clean figures wherever required.
10. Take π =22/7 wherever required if not stated.
11. Use of calculators is not allowed.

SECTION-A

This section consists of 20 questions of 1 mark each.

Q.1 Find the H.C.F. of the 336 and 54.

(a) 6 (b) 10 (c) 12 (d) 18

Q.2 Find the zeroes of the quadratic polynomial $x^2+7x+10$

(a) $-1,-4$ (b) $-2,-5$ (c) $-1,-7$ (d) None of these

Q.3 If one zero of the quadratic polynomial x^2+7x+k is 2,then the value of k is

(a) 10 (b) -10 (c) 5 (d) -5

Q.4 A Polynomial of degree zero is

(a) Linear (b) Quadratic (c) Constant (d) Cubic

Q.5 If a pair of linear equations is consistent, then the lines are:

(a) Parallel (b)Always coincident (c) Always Intersecting (d) Intersecting or coincident

Q.6 If one root of equation $4x^2-2x+k-4 = 0$ is reciprocal of the other. The value of k is:

(a) 8 (b) -8 (c) 4 (d) -4

Q.7 30^{th} term of the A.P: 10,7, 4,……… is

(a) 97 (b) 77 (c) -87 (d) -77

Q.8 The midpoint of a line segment joining two points A(2, 4) and B(-2, -4) is

(a) (-2,4) (b) $(2,-4)$ (c) (0,0) (d) (-2,-4)

Q.9 The point which divides the line segment of points P(-1, 7) and (4, -3) in the ratio of 2:3 is:

(a) (-1,3) (b) (-1,-3) (c) (1,-3) (d) (1,3)

Q.10 The distance between the points (0, 5) and (–5, 0) is

(a) 5 unit (b) $5\sqrt{2}$ unit (c) $2\sqrt{5}$ unit (d) 10 unit

Q.11 If TP and TQ are the two tangents to a circle with centre O so that ∠POQ = 110°, then ∠PTQ is equal to

(a) 60° (b) 70° (c) 80° (d) 90°

Q.12 In ∆ ABC, right-angled at B, AB = 24 cm, BC = 7 cm. The value of tan C is:
(a) $12/7$ (b) 20/7 (c) 7/24 (d) 24/7

Q.13 The value of $sin^2 60° - cos^2 30° + 2tan 45°$ is

(a) 3 (b) 1 (c) 2 (d) 0

Q.14 If we join two hemispheres of same radius along their bases, then we get a;

(a) Cone (b) Cylinder (c) Sphere (d) Cuboid

Q.15 Volumes of two spheres are in the ratio 64:27. The ratio of their surface areas is:

(a) 3:4 (b) 4:3 (c) 9:16 (d) 16:9

Q.16 If the Median and Mean of a distribution are 25 and 30 respectively, then the Mode of the distribution is:

(a) 15 (b) 17 (c) 25 (d) 20

Q.17 Consider the following frequency distribution

Class	0-5	6-11	12-17	18-23	24-29
Frequency	13	10	15	8	11

The upper limit of the median class is

(a) 17 (b) 17.5 (c) 18 (d) 18.5

Q.18 If we throw two coins in the air, then the probability of getting both tails will be:

(a) 1/2 (b) 0 (c) 1/4 (d) None of these

Directions: In question number 19 and 20 a statement of Assertion (A) is followed by a statement of Reason(R) Choose the correct option.
(a) Both, Assertion(A) and Reason(R) are true and Reason(R) is correct explanation of Assertion (A)
(b) Both, Assertion(A) and Reason(R) are true and Reason(R) is not correct explanation for Assertion (A)
(c) Assertion(A) is true but Reason(R) is false.
(d) Assertion(A) is false but Reason(R) is true.

Q.19 **Assertion(A):** The probability of winning a game is 0.4, then the probability of losing it, is 0.6.

Reason(R): P(E) + P (not E) = 1

Q.20 **Assertion(A)**: Two dice are thrown simultaneously. There are 11 possible outcomes and each of them has a probability 1/11.

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Reason(R): Probability P(E) = $\frac{Number\ of\ favourable\ outcomes}{Total\ number\ of\ possible\ outcomes}$

SECTION-B

This section consists of 5 questions of 2 marks each.

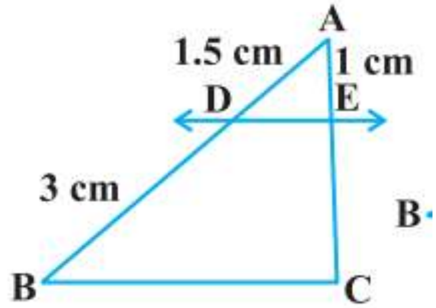
Q.21 Find the HCF and LCM of 6, 72 and 120, using the prime factorisation method

OR

Explain why 7 × 11 × 13 + 13 and 7 × 6 × 5 × 4 × 3 × 2 × 1 + 5 are composite numbers.

Q.22 Solve the pair of linear equations $x + y = 14$ and $x - y = 4$ by the substitution method.

Q.23 In the Figure given below , DE || BC. Find EC



OR

E and F are points on the sides PQ and PR respectively of a Δ PQR. State whether EF || QR :
PE = 3.9 cm, EQ = 3 cm, PF = 3.6 cm and FR = 2.4 cm

Q.24 If sinA = $\frac{3}{4}$, calculate cos A and tan A

Q.25 One card is drawn from a well-shuffled deck of 52 cards. Calculate the probability that the card will (i) be an ace, (ii) not be an ace.

SECTION-C

This section consists of 6 questions of 3 marks each.

Q.26 Prove that $5 - \sqrt{2}$ is an irrational number, If we already know that √2 is irrational.

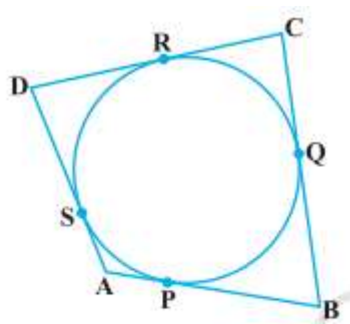
Q.27 Five years ago, Nuri was thrice as old as Sonu. Ten years later, Nuri will be twice as old as Sonu. How old are Nuri and Sonu?

Q.28 Find the coordinates of the point which divides the line segment joining the points (4, – 3) and (8, 5) in the ratio 3: 1 internally

OR

Check whether (5, – 2), (6, 4) and (7, – 2) are the vertices of an isosceles triangle.

Q.29 A quadrilateral ABCD is drawn to circumscribe a circle. Prove that AB + CD = AD + BC



Q.30 Prove that $\frac{cosA}{1+sinA} + \frac{1+\sin A}{cosA}$ = 2sec A

OR

Prove that $(\sin\theta + \text{cosec}\,\theta)^2 + (\cos\theta + \sec\theta)^2 = 7 + \tan^2\theta + \cot^2\theta$

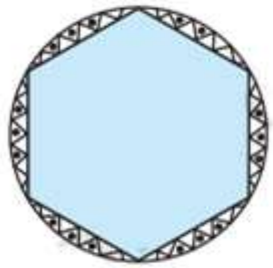
Q.31 A vessel is in the form of an inverted cone. Its height is 8 cm and the radius of its top, which is open, is 5 cm. It is filled with water up to the brim. When lead shots, each of which is a sphere of radius 0.5 cm are dropped into the vessel, one-fourth of the water flows out. Find the number of lead shots dropped in the vessel.

SECTION-D

This section consists of 4 questions of 5 marks each.

Q.32 A 1.5 m tall boy is standing at some distance from a 30 m tall building. The angle of elevation from his eyes to the top of the building increases from 30° to 60° as he walks towards the building. Find the distance he walked towards the building.

Q.33 A round table cover has six equal designs as shown in figure. If the radius of the cover is 28 cm, find the cost of making the designs at the rate of Rs 0.35 per sq cm. (Use $\sqrt{3} = 1.7$)



Q.34 Which term of the A.P. 3, 15, 27, 39,.. will be 132 more than its 54^{th} term?

OR

Find the sum of first 51 terms of an AP whose second and third terms are 14 and 18 respectively.

Q.35 State and prove Basic Proportionality Theorem) (BPT).

OR

Sides AB and BC and median AD of a triangle ABC are respectively proportional to sides PQ and QR and median PM of Δ PQR (see Fig. 6.41). Show that Δ ABC ~ Δ PQR

SECTION-E

This section consists of 3 Case –Study Based Questions of 4 marks each.

Q.36 Traffic Management: A traffic enforcement camera is a camera which may be mounted beside or over a road or installed in an enforcement vehicle to detect motoring offenses, including speeding, vehicles going through a red traffic light. A worldwide review of studies found that speed cameras led to a reduction of 11% to 44% for fatal and serious injury crashes. The British Medical Journal recently reported that speed cameras were effective at reducing accidents and injuries in their vicinity and recommended wider deployment.:

Speed (km/h)	Number of Vehicles
20-30	1
30-40	3
40-50	7
50-60	16
60-70	35
70-80	29
80-90	7
90-100	2

Based on the above information, answer the following questions:

(i). Find the number of vehicles whose speed is more than 70 km/h and find the number of vehicles whose speed is less than 50 km/h ? (1 mark)

(ii) What is the class mark of the modal class? (1 mark)

(iii) Find the mode of the given data. 2 marks)

OR

Find median of the given data.

Q.37 Ritu has a lawn with a flowerbed and grass land. The grass land is in the shape of rectangle while flowerbed is in the shape of square. The length of the grassland is found to be 3 m more than twice the length of the flowerbed. Total area of the whole lawn is 1260 sq m.

Based on the above information, answer the following questions:

(i) If the length of the flowerbed is x m then what is the total length of the lawn ? (1 mark)

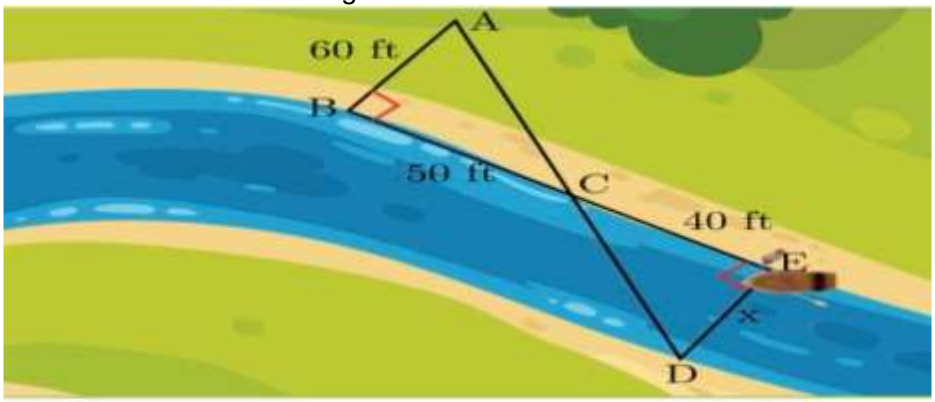
(ii) What is the value of x if the area of total lawn is 1260 m^2 ? (1 mark)

(iii) What is the area of grassland ? (2 marks)

OR

What is the ratio of area of flowerbed to area of grassland ?

Q.38 Tania is very intelligent in maths. She always try to relate the concept of maths in daily life. One day she plans to cross a river and want to know how far it is to the other side. She takes measurements on her side of the river and make the drawing as shown below



Based on the above information, answer the following questions:

(i) Which similarity criterion is used in solving the above problem ? (1 mark)

(ii) Consider the following statement : (1 mark)

$S1$: $\angle ACB = \angle DCE$

$S2$: $\angle BAC = \angle CDE$

Which of the above statement is/are correct.

(a) $S1$ and $S2$ both (b) $S1$

(c) $S2$ (d) None

(iii) Consider the following statement : (2 mark)

(A) $\frac{AB}{DE} = \frac{CA}{CD}$ (B) $\frac{BC}{CE} = \frac{AB}{DE}$

(C) $\frac{CA}{CD} = \frac{DE}{AB}$

Which of the above statements are correct ?

(a) A and C (b) B and C

(c) A and B (d) All three

Or

What is the approximate length of ED shown in the figure?